

JustIT

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Ingegneria del Software e Progettazione Web

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1. Software Requirement Specification

1.1. Introduction

1.1.1. Aim of the document

The purpose of this document is to present the fundamental and specific requirements of the “JustIT” system, a project for the “ISPW” Academic Year of 2025-2026

The document provides a comprehensive overview of the system’s objective core functionalities and the rationale behind the design choices, in order to guide the development process and ensure alignment among all team members.

Team members: Valerio Mazza, Giulio Rustia

1.1.2. Overview of the defined system

The system is a Java-based software platform structured according to the MVC architecture developed with Maven, and featuring two user interfaces:

- JavaFX and FXML files.
- Standard Command Line Interface.

The system enables efficient booking and appointment management for technical shops. Features included are:

- Management of clients and technicians.
- Management of technical shops.
- Browsing and searching technical shops near you.
- A system to review the service.
- View management of the bookings and appointment
- Exporting a booking calendar on csv

The system supports two types of users, each with different permissions and functionalities:

- Technicians.
- Customers.

Furthermore, the application can be executed in two different modes:

- Demo-version: execution with in-memory persistence (volatile data).
- Full-version: execution with persistence on a relational database or file system.

The system supports two persistence modes:

- Relational SQLite database.
- File-System based on Json files.

1.1.3. HW e SW requirements

Software requirements

- Java 21+
- IDE: IntelliJ IDEA or Eclipse
- SQLite Database
- Operating System: Any Linux based distro with java 21+ support, Windows 10+, macOS 10.12+

Hardware Requirements

- CPU: Dual-Core 2.0 Ghz
- Ram: 2 GB

- Disk space: 1 GB

1.1.4. Related systems

1.1.4.1. TaskRabbit

Pro

- Extensive directory of professionals with category filters
- Robust review and reputation system
- Detailed requests with price quotes

Cons

- Workflow is too fragmented for quick bookings

1.1.4.2. ProntoPro

Pro

- Prioritizes local pros and small businesses
- Easy quote comparison
- Covering all of Italy

Cons

- Less emphasis on map/radius search

1.2. User Stories

1.2.1. User

- As a User, I want to find all shops near me, so that i can choose which one to call.
- As a User, I want to filter technicians by distance, so that I can receive support faster
- As a User, I want to rate and review my experience, so that I can share feedback on the shop's service quality

1.2.2. Technician

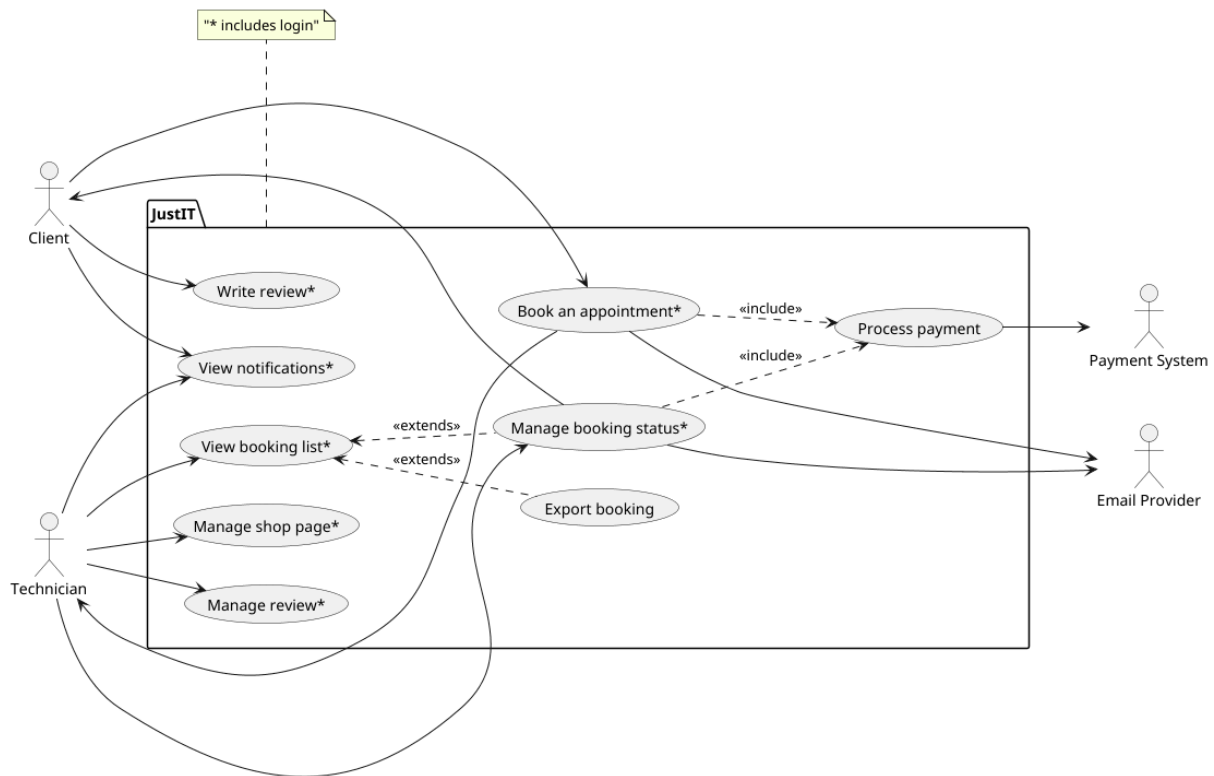
- As a Technician, I want to accept or reject appointments, so that I can ensure I only take jobs within my area of expertise.
- As a Technician, I want to list my service offerings, so that potential clients can easily view what I provide.
- As a Technician, I want to download my booking history, so that I can manage my agenda more efficiently.

1.3. Functional Requirements

- The system shall display the current status of a booking.
- The system shall notify the User of any booking status change
- The system shall provide a filter for Users to display shops located within a customizable radius from their address location
- The system shall display a detailed profile for each Shop, including services and reviews.
- The system shall provide the Technician's booking history in CSV format.
- The system shall provide a booking form for the User to describe the issue.

1.4. Use Cases

1.4.1. Use Cases Diagram



1.4.2. Use case internal steps

1.4.2.1. Use case: Book an appointment

Main flow:

- 1. The user accesses the “Browse Shop” section via Login.
- 2. The system displays a selection of nearby shops.
- 3. The user chooses a shop.
- 4. The system displays the shop page information.
- 5. The user selects “Book Now”.
- 6. The system shows a booking form.
- 7. The user selects the preferred time slot and confirms the selection.
- 8. The system processes the payment, records the booking and notifies the technician.

Extensions:

- 1a. The user fails the authentication.
- 1b. The system displays an error and allows them to retry.
- 7a. The user has already booked the selected shop.
- 7b. The system displays an error message and returns the user to the Shop page.
- 8a. The user fails to complete the payment within 5 minutes.
- 8b. The system display an error message and returns the user to the booking form.

1.4.2.2. Use case: Manage Booking Status

Main flow:

- 1. The technician accesses the “Booking List” section.

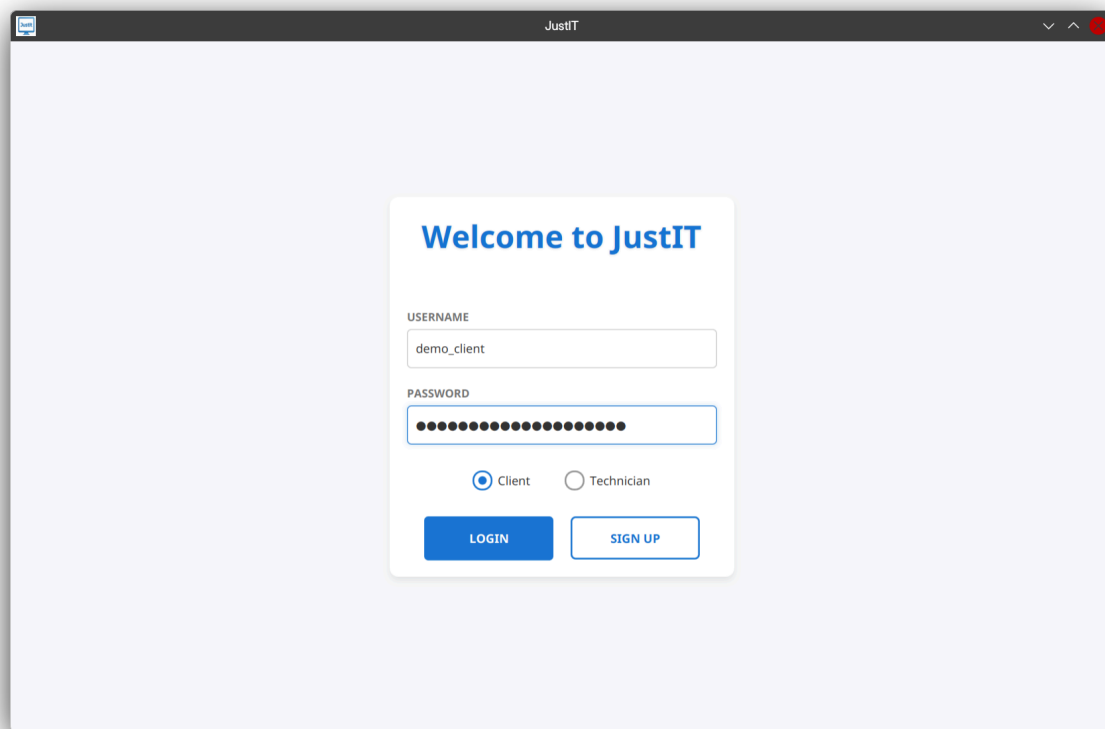
- 2. The system displays the list of bookings.
- 3. The technician selects a booking.
- 4. The system displays the details of booking.
- 5. The technician changes the status of the booking.
- 6. The system validates the requested status transition.
- 7. The system updates the booking status.
- 8. The system sends a notification to the user associated with the booking.

Extension:

- 2a. The system detects that there are no bookings yet.
- 2b. The system displays a message informing the Technician that no bookings are available.
- 6a. The technician selects a status change that is not allowed.
- 6b. The system prevents the status update.
- 6c. The system displays an error message indicating that the selected transition is not valid.

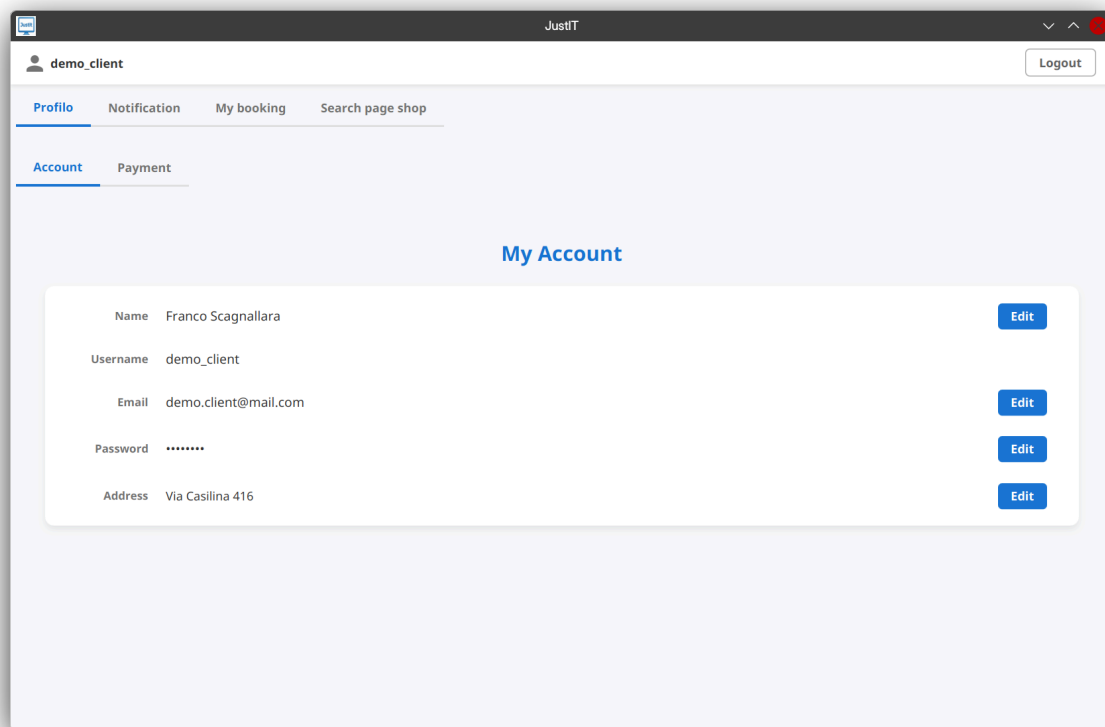
2. Storyboards

2.1. Login

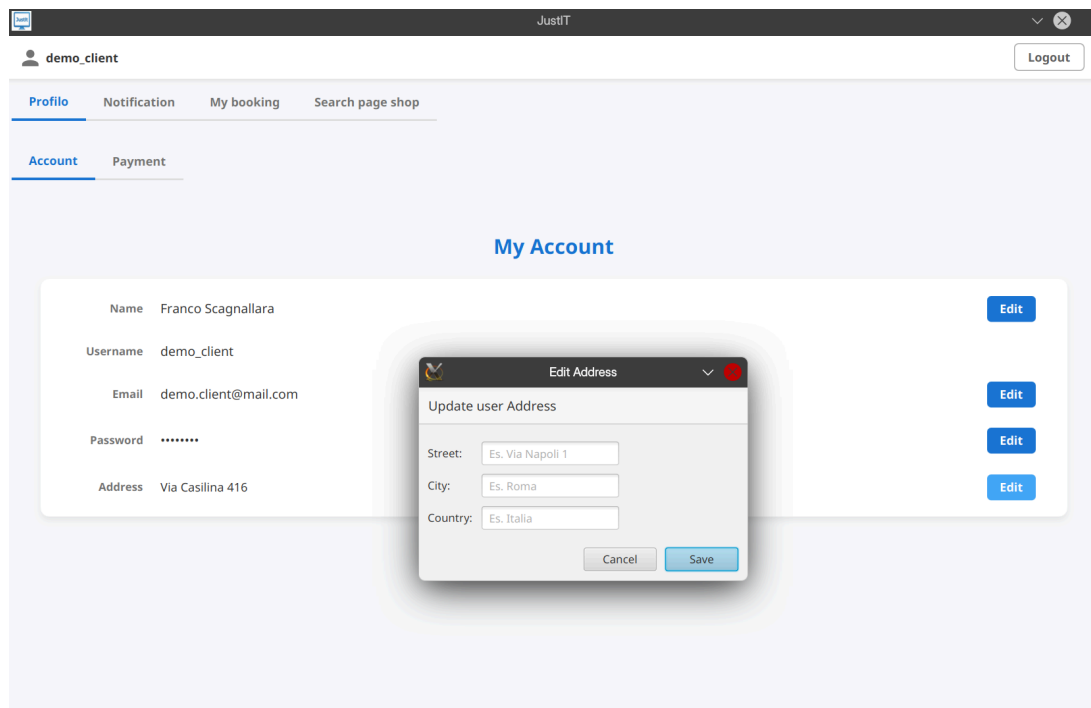


The screenshot shows a web browser window titled "JustIT". The main content area is light blue. In the center, there is a white login card with a blue header "Welcome to JustIT". Below the header, there are two input fields: "USERNAME" with the text "demo_client" and "PASSWORD" with a masked password represented by black dots. Below the password field, there are two radio buttons: "Client" (selected) and "Technician". At the bottom of the card, there are two buttons: "LOGIN" (blue) and "SIGN UP" (white with a blue border).

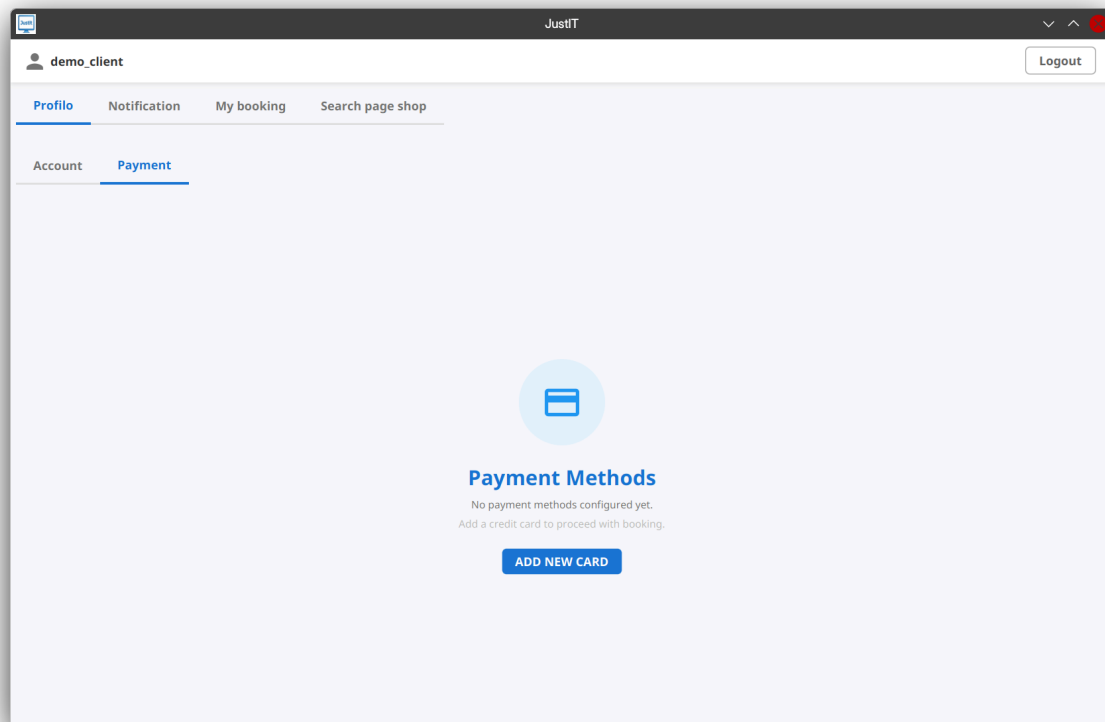
2.2. Account page client



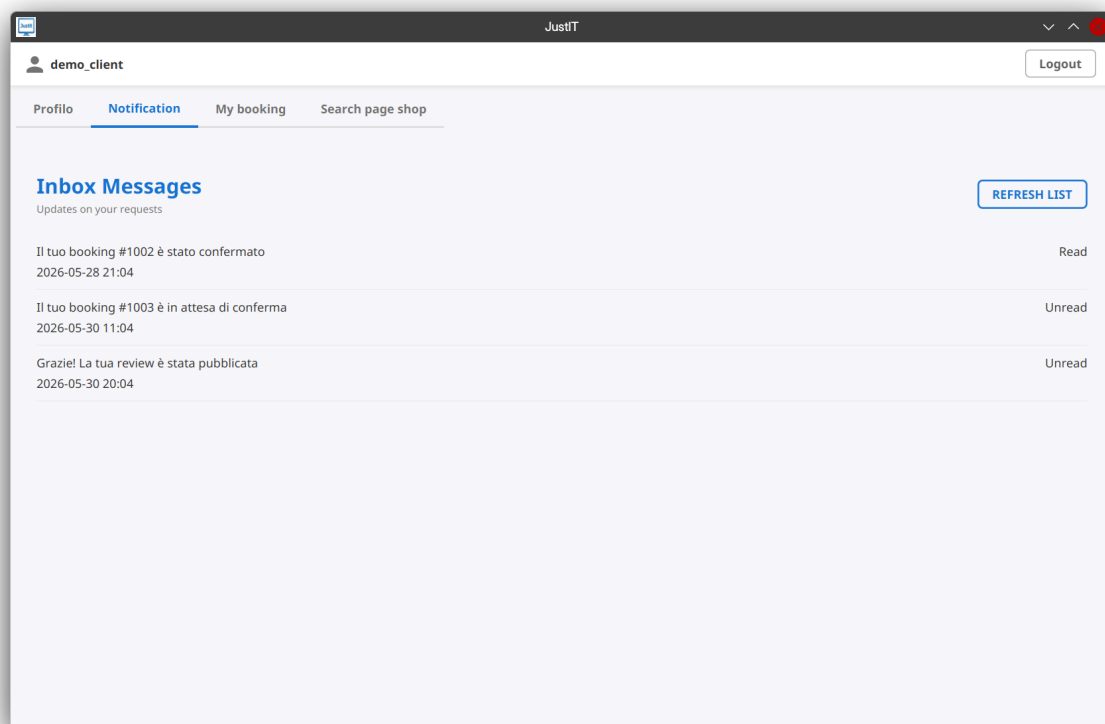
2.3. Update address



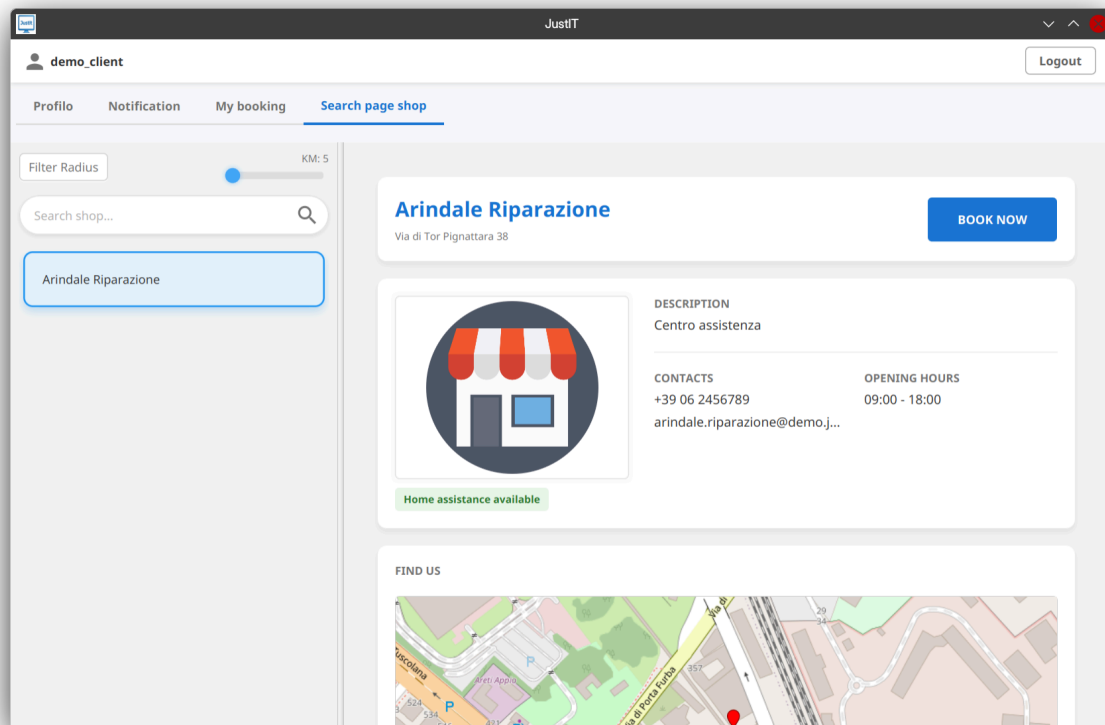
2.4. Add payment card



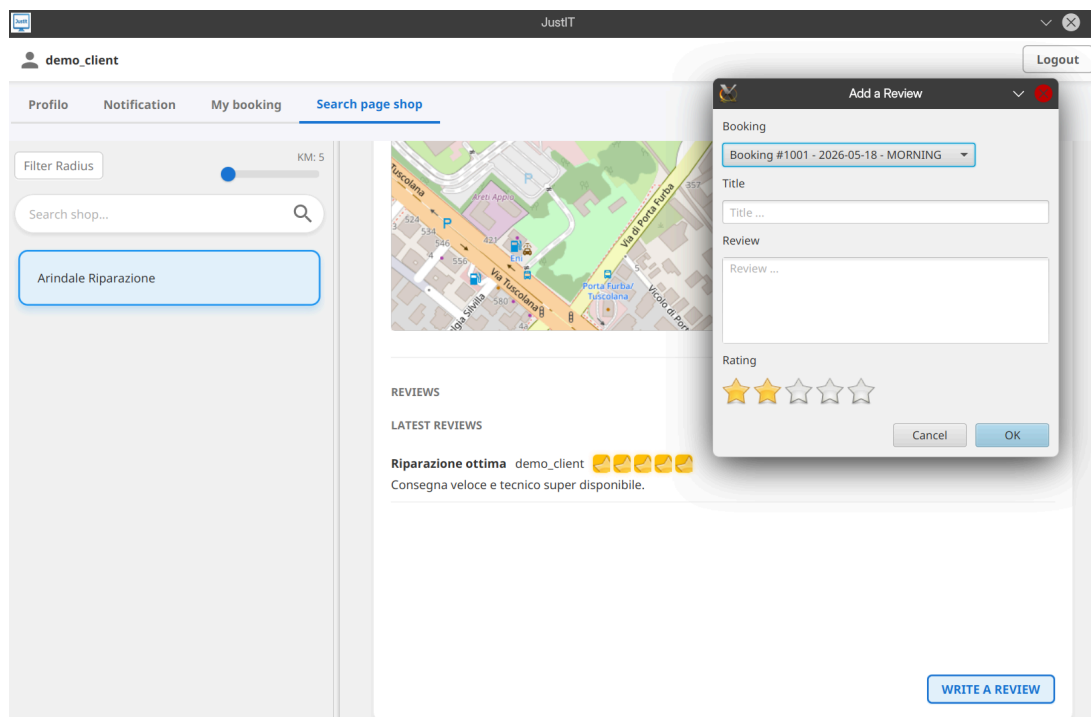
2.5. Inbox notification client



2.6. Search and Page shop



2.7. Write review



2.8. Booking

The screenshot shows the 'Book Appointment' form in the JustIT application. The user is logged in as 'demo_client'. The navigation bar includes 'Profile', 'Notification', 'My booking', and 'Search page shop'. The left sidebar has a 'Filter Radius' slider set to 5 KM and a search bar with 'Arindale Riparazione' selected. The main form area is titled 'Book Appointment' with the subtitle 'Schedule a visit with the shop'. It contains the following fields and options:

- PICK A DATE:** A date picker showing '6/7/2026'.
- SELECT TIME SLOT:** A dropdown menu showing 'MORNING'.
- ADDITIONAL SERVICES:** A checkbox labeled 'Request home visit' which is checked.
- DESCRIBE THE PROBLEM:** A text area with the placeholder 'Describe briefly your issue...'.
- CONFIRM BOOKING:** A blue button at the bottom of the form.

2.9. Booking list

The screenshot shows the 'My Bookings' list in the JustIT application. The user is logged in as 'demo_client'. The navigation bar includes 'Profile', 'Notification', 'My booking', and 'Search page shop'. The left sidebar has a 'Filter Radius' slider set to 5 KM and a search bar with 'Arindale Riparazione' selected. The main form area is titled 'My Bookings' with the subtitle 'Track the status of your appointments'. It contains a list of three bookings:

Shop	Date	Time	Status	Description	Home Assistance
Arindale Riparazione	2026-05-18	MORNING	COMPLETED	Sostituzione batteria Stonex One	No
Arindale Riparazione	2026-05-28	AFTERNOON	CONFIRMED	Installazione sailfish os	Yes
Arindale Riparazione	2026-06-02	EVENING	PENDING	Pulizia steam controller	No

2.10. Account tech

The screenshot shows the 'My Account' page of the JustIT application. The user is logged in as 'demo_tech'. The page has a navigation bar with 'PageShop', 'Booking list', 'Notification', 'Profile' (active), and 'Review'. The main content area is titled 'My Account' and contains a form with the following fields:

Field	Value	Action
Name	Giulio Agricolo	Edit
Username	demo_tech	
Email	demo.tech@mail.com	Edit
Password	*****	Edit

2.11. Manage bookings

The screenshot shows the 'Booking Requests' page of the JustIT application. The user is logged in as 'demo_tech'. The page has a navigation bar with 'PageShop', 'Booking list' (active), 'Notification', 'Profile', and 'Review'. The main content area is titled 'Booking Requests' and contains a table of booking requests. There is also an 'EXPORT CSV' button.

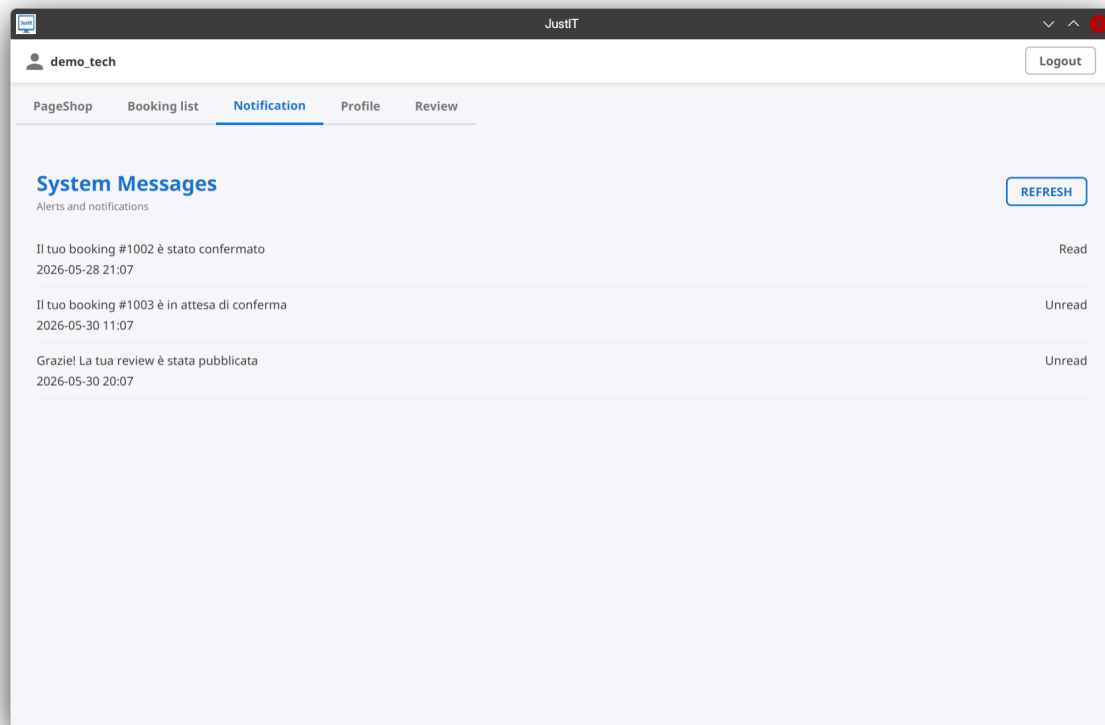
User	Date	Time Slot	Status	Home Assist.
demo_client	2026-05-18	MORNING	COMPLETED	No
demo_client	2026-05-28	AFTERNOON	CONFIRMED	Yes
demo_client	2026-06-02	EVENING	PENDING	No
demo_client_2	2026-05-25	MORNING	REJECTED	No

Below the table, there is a 'BOOKING DETAILS' section with the following information:

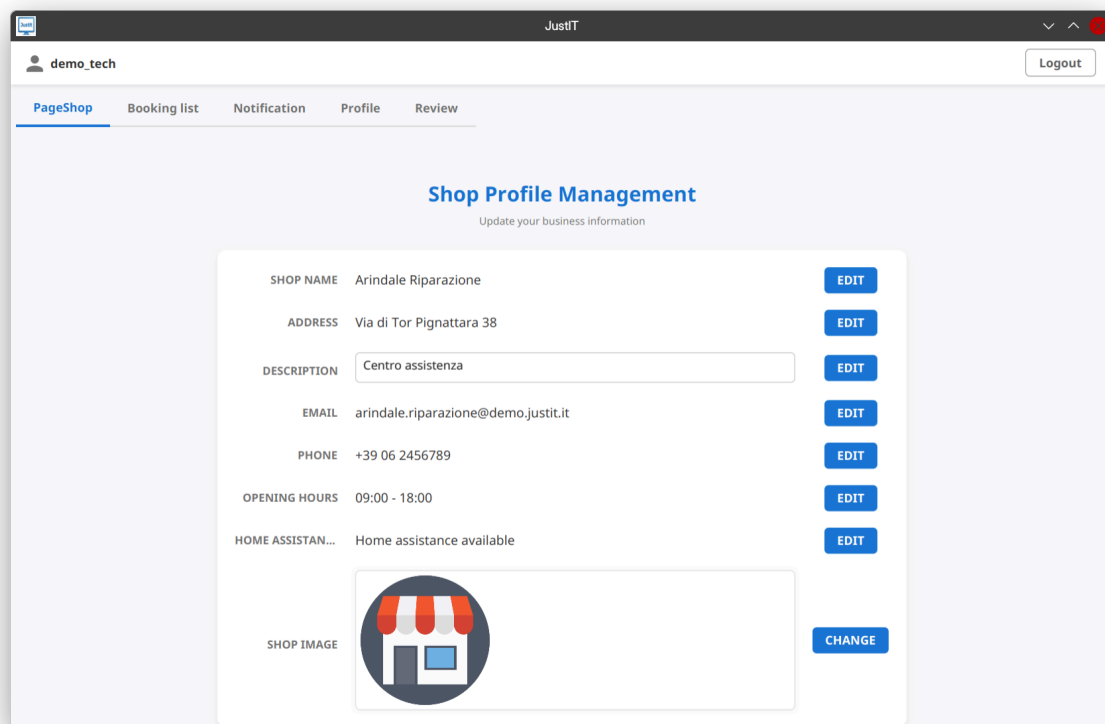
Field	Value
CLIENT:	User: demo_client
DATE & TIME:	Date: 2026-06-02 @ Time Slot: EVENING
ADDRESS:	
STATUS:	Status: PENDING
SERVICE:	Home Assistance: No

Below the details, there is a 'PROBLEM DESCRIPTION' section with a text area containing the text 'Pulizia steam controller'. At the bottom right, there are two buttons: 'REJECT' and 'APPROVE REQUEST'.

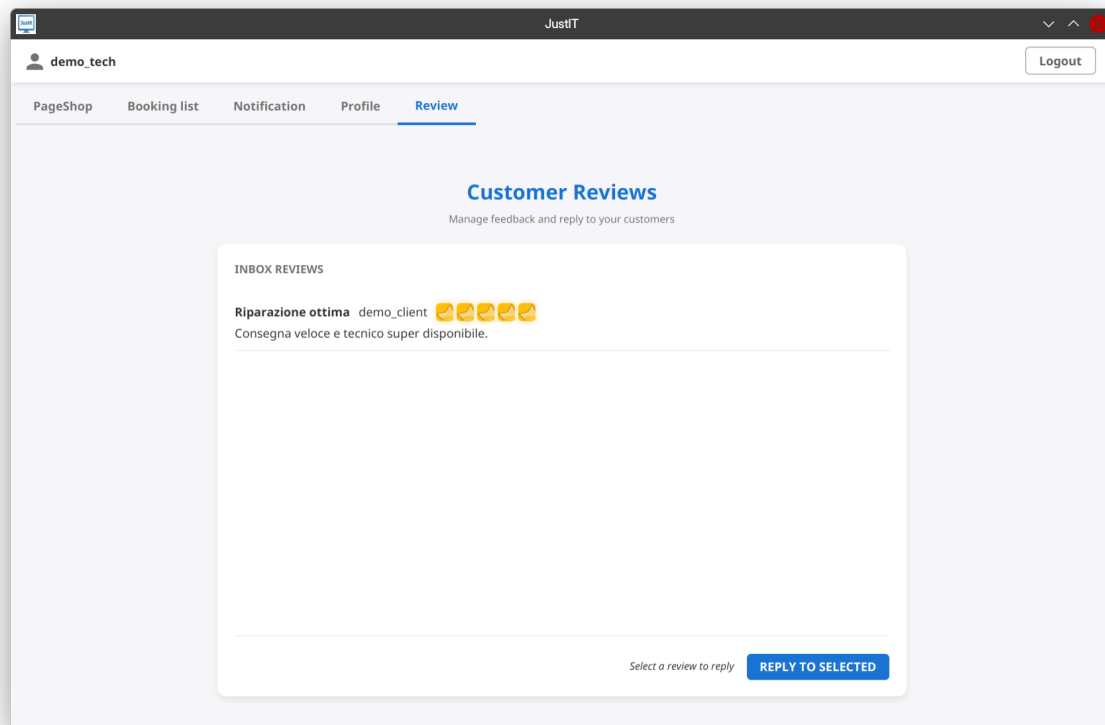
2.12. System notification



2.13. Manage page shop



2.14. Manage customer reviews



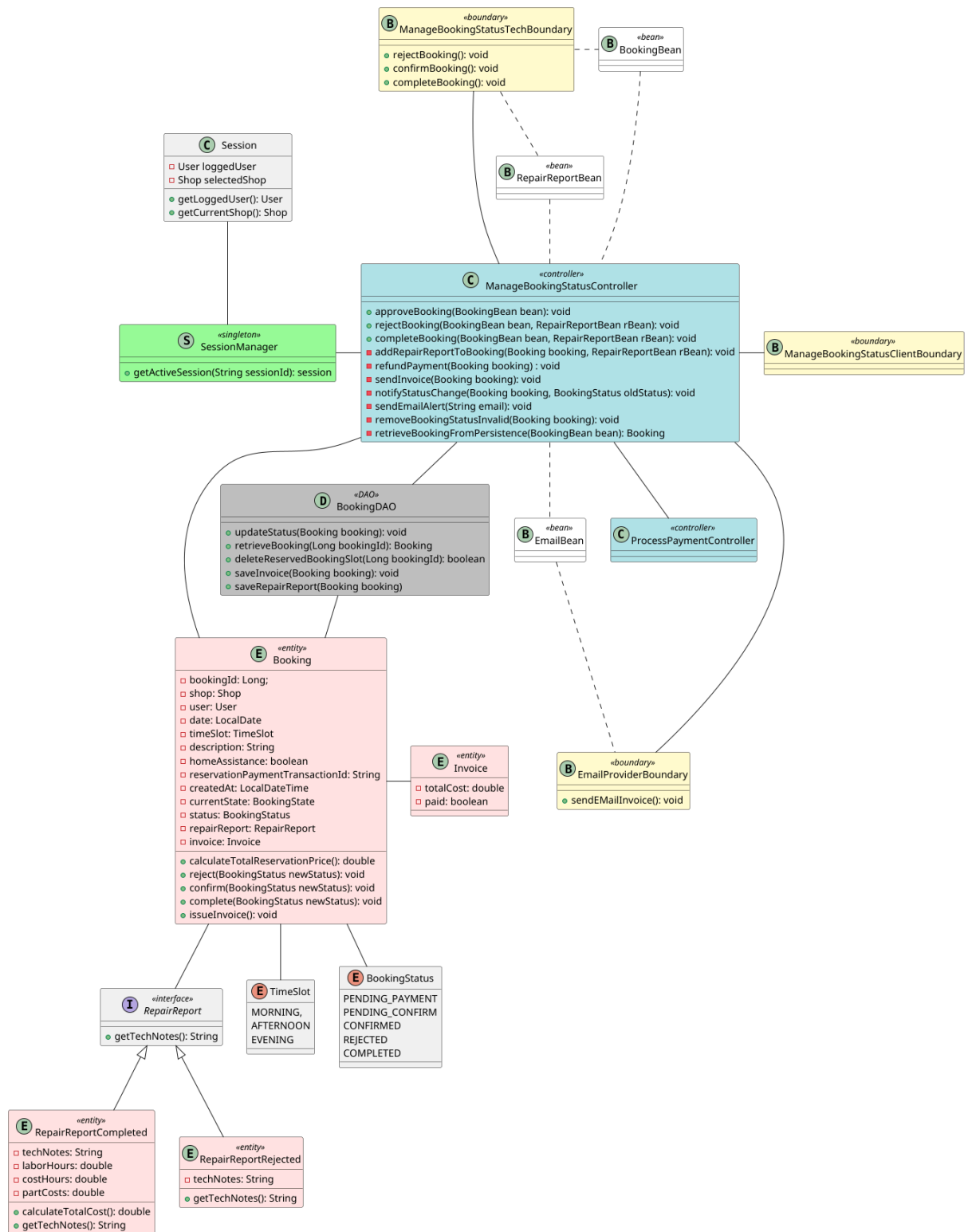
3. Design

Since design diagrams are created before development begins, they may not perfectly reflect the final codebase.

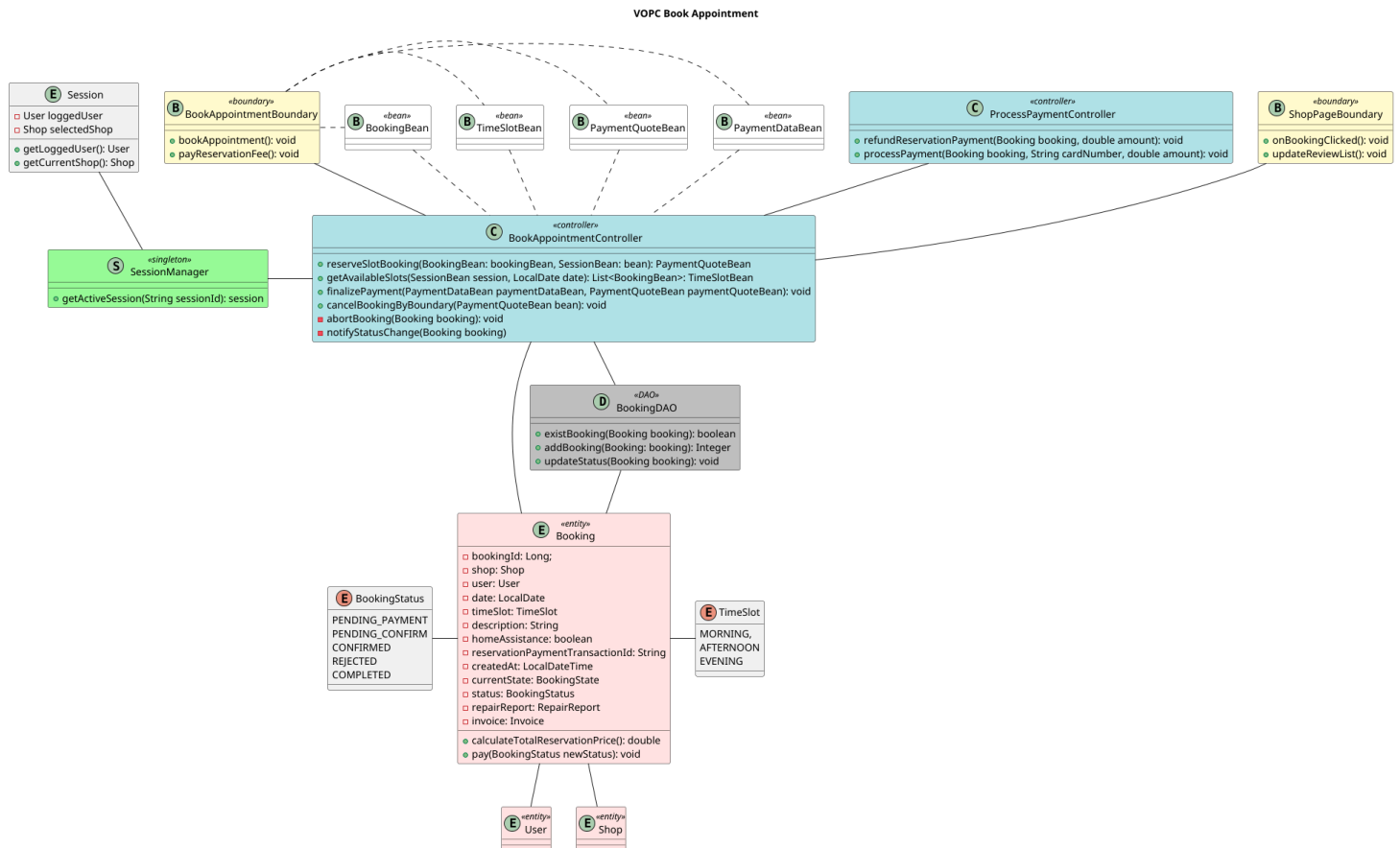
3.1. Class diagram

3.1.1. VOPC Analysis

3.1.1.1. Manage Booking Status

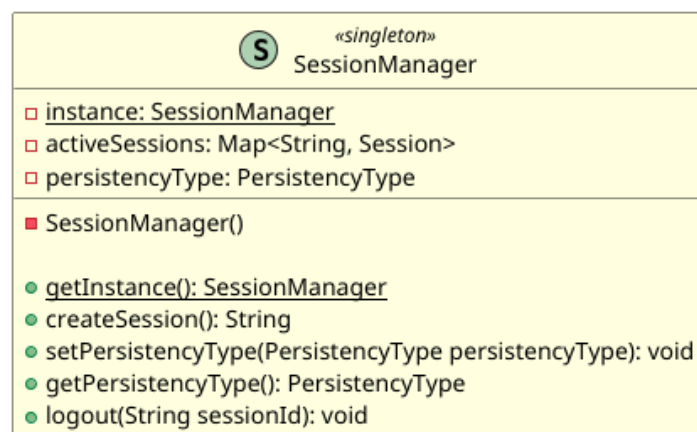


3.1.1.2. Book Appointment

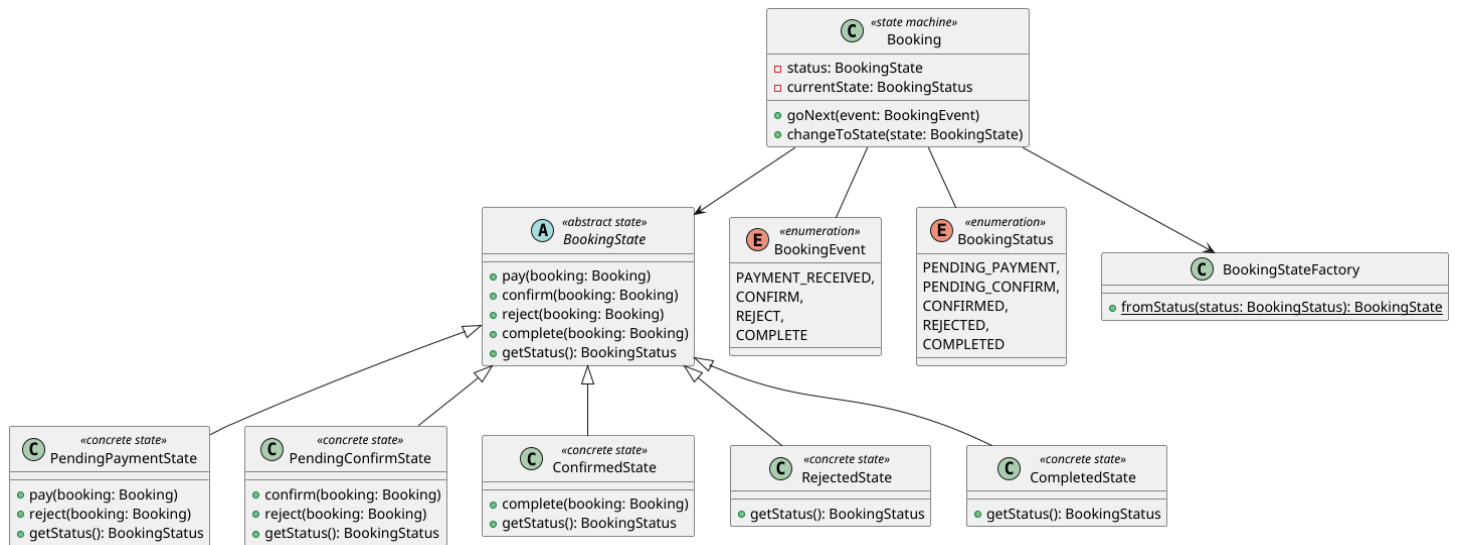


3.1.2. Design Level Diagram

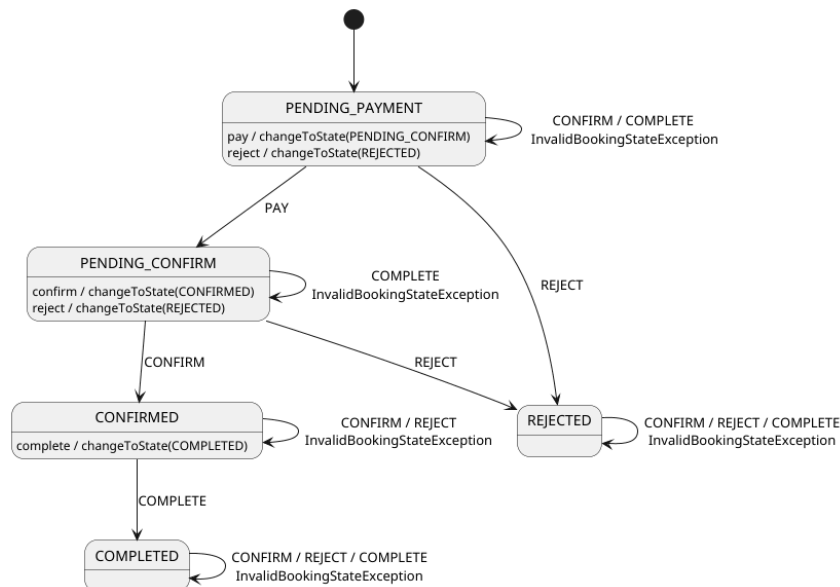
3.1.2.1. Singleton



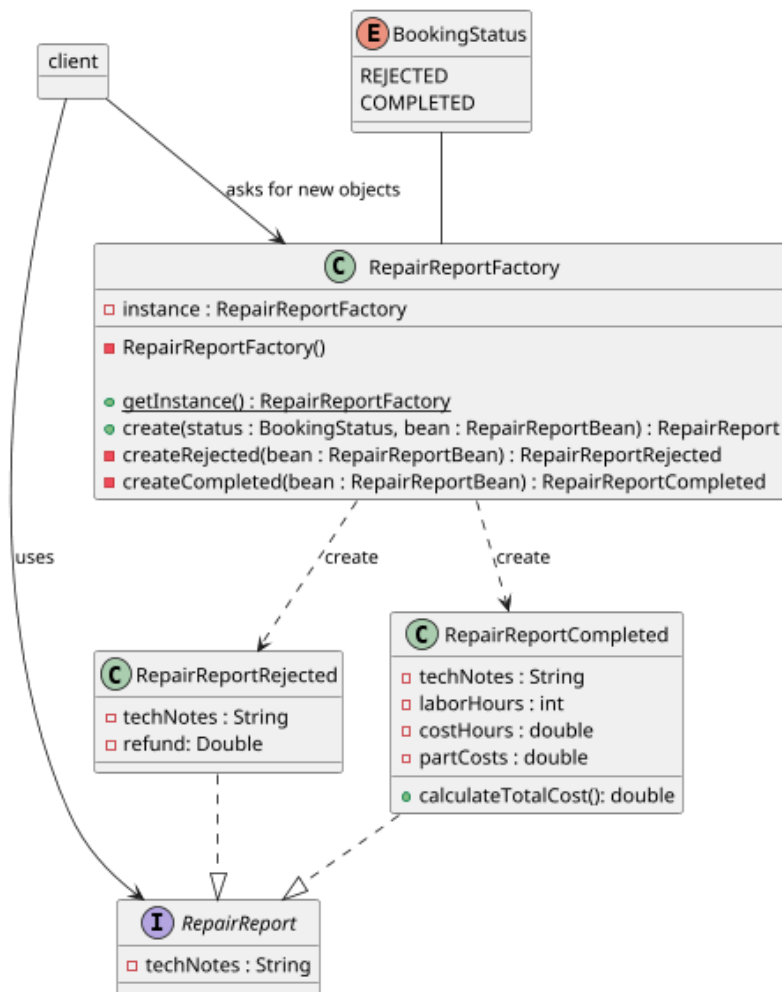
3.1.2.2. State Pattern



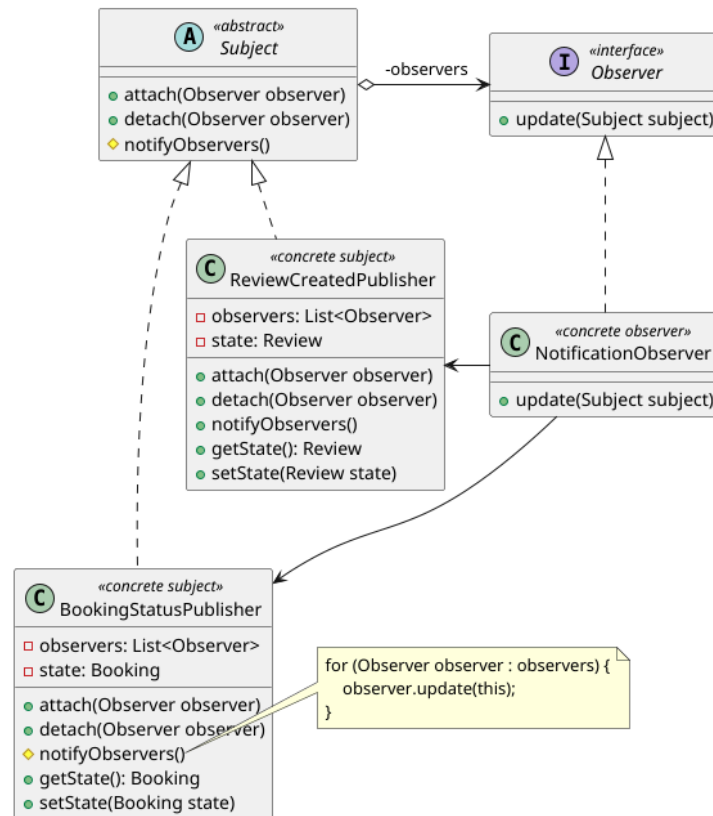
3.1.2.2.1. State Machine Diagram



3.1.2.3. Factory Method



3.1.3. Observer



3.2. Design patterns

3.2.1. Singleton pattern

In JustIT, the Singleton pattern is employed to manage user sessions through a centralized SessionManager. This component acts as the single point of access for the application's state, globally accessible via getInstance(). This architectural choice prevents the anti-pattern of passing session objects repetitively throughout the system, thereby ensuring data consistency and low coupling.

Upon a successful login, the controller generates a new Session object encapsulating the LoggedUser instance. This session is securely registered within the SessionManager's internal map using a unique UUID as a key. By invoking getActiveSession(sessionId), any layer of the application can securely retrieve user data (getLoggedUser) and perform Role-Based Access Control checks (e.g., isClient, isTechnician).

Additionally, the session dynamically handles the context of the current shop via setCurrentShop and getCurrentShop. For Clients, it stores the shop they are currently browsing, whereas for Technicians, it stores their affiliated shop. This provides a Single Source of Truth, allowing the Booking, Review, Notification, and Account controllers to reliably access shop details without duplicating business logic.

Finally, the logout operation explicitly invalidates the UUID and clears the associated shared state from the manager, preventing memory leaks and ensuring that no sensitive information remains accessible after the session ends.

3.2.2. State Pattern

In JustIT, the State pattern is applied to manage the complex lifecycle of a Booking. This pattern localizes and partitions state-specific behaviors into distinct concrete classes (e.g., PendingPaymentState, ConfirmedState) that implement a common BookingState interface. Each state transition and its

associated business logic is encapsulated within its respective class, elevating the execution state to full object status. This structure makes the intent of the code clearer: default illegal operations are handled safely by the interface, and valid transitions are explicitly localized, ensuring that the system can be easily extended with new states without altering the Booking context.

3.2.3. Factory Method Pattern

In JustIT, the Factory Method pattern is adopted to manage the polymorphic creation of repair reports through the centralized RepairReportFactory class.

The factory is responsible for providing the appropriate concrete implementation of a repair report based on the outcome of a booking. The selection process is driven by the BookingStatus passed to the create() method alongside the data payload (RepairReportBean). If the provided BookingStatus is not supported by the factory, an exception is thrown, preventing the creation of invalid repair report instances.

The RepairReport interface defines the baseline contract. Concrete implementations encapsulate specific business rules and state: RepairReportCompleted manages metrics such as labor hours, part costs, and total cost calculation, whereas RepairReportRejected handles scenarios like refund management.

This design decouples the client (the Application Controller) from the concrete instantiation logic. The client interacts exclusively with the RepairReportFactory and the RepairReport abstraction, remaining entirely unaware of the specific subclass being instantiated.

Centralizing object creation in the factory improves modularity, maintainability, and scalability, enabling the addition of new types of repair report implementations with minimal impact on the existing codebase.

3.2.4. Observer pattern

In JustIT, the Observer pattern is used to handle notifications between components without creating direct dependencies between them.

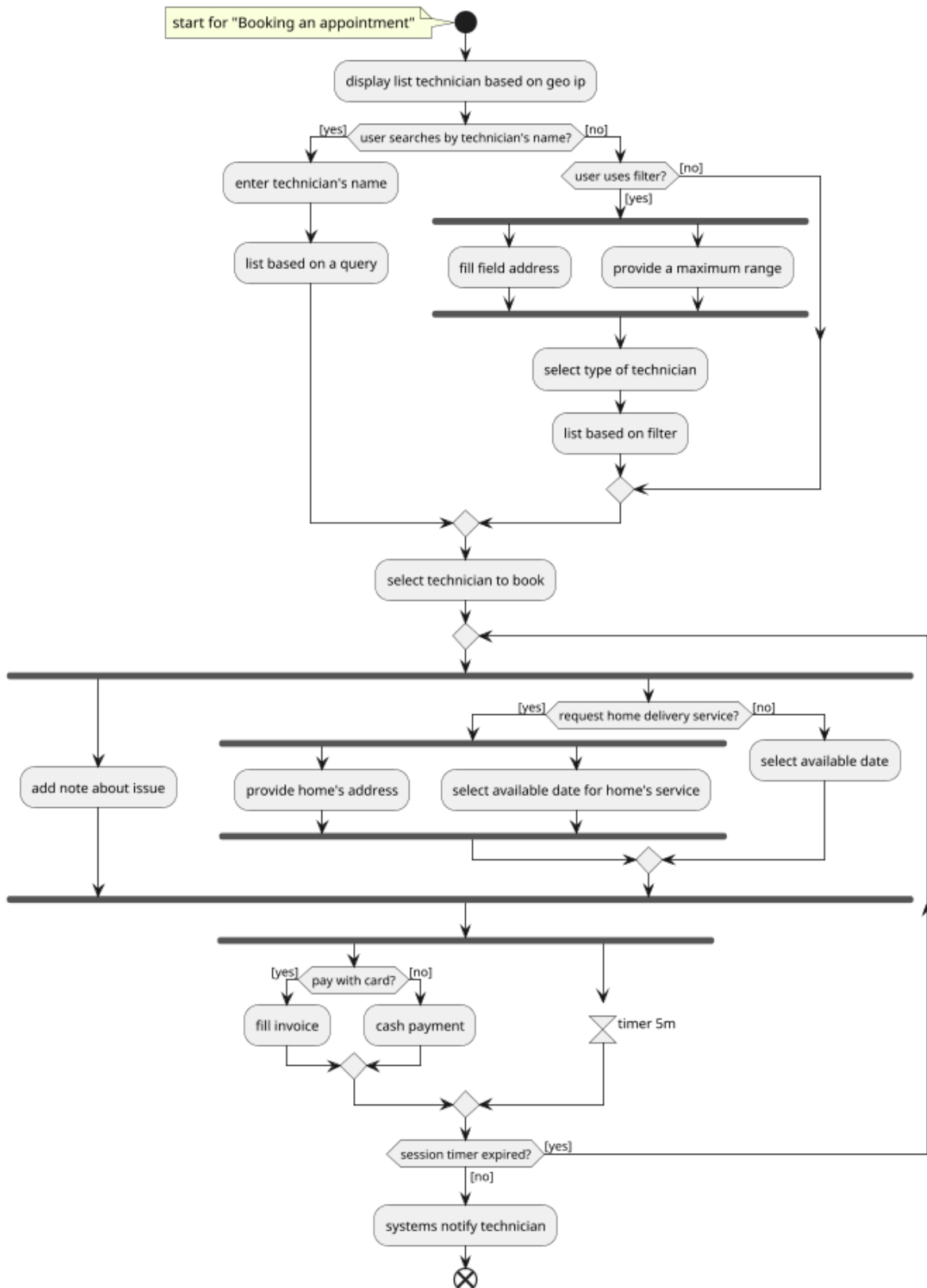
This pattern is crucial for maintaining state consistency between core business entities (such as Bookings and Reviews) and the notification subsystem, without tightly binding them.

We introduced an abstract class Subject and an Observer interface. Specialized singleton classes, BookingStatusPublisher and ReviewCreatedPublisher, act as Concrete Subjects.

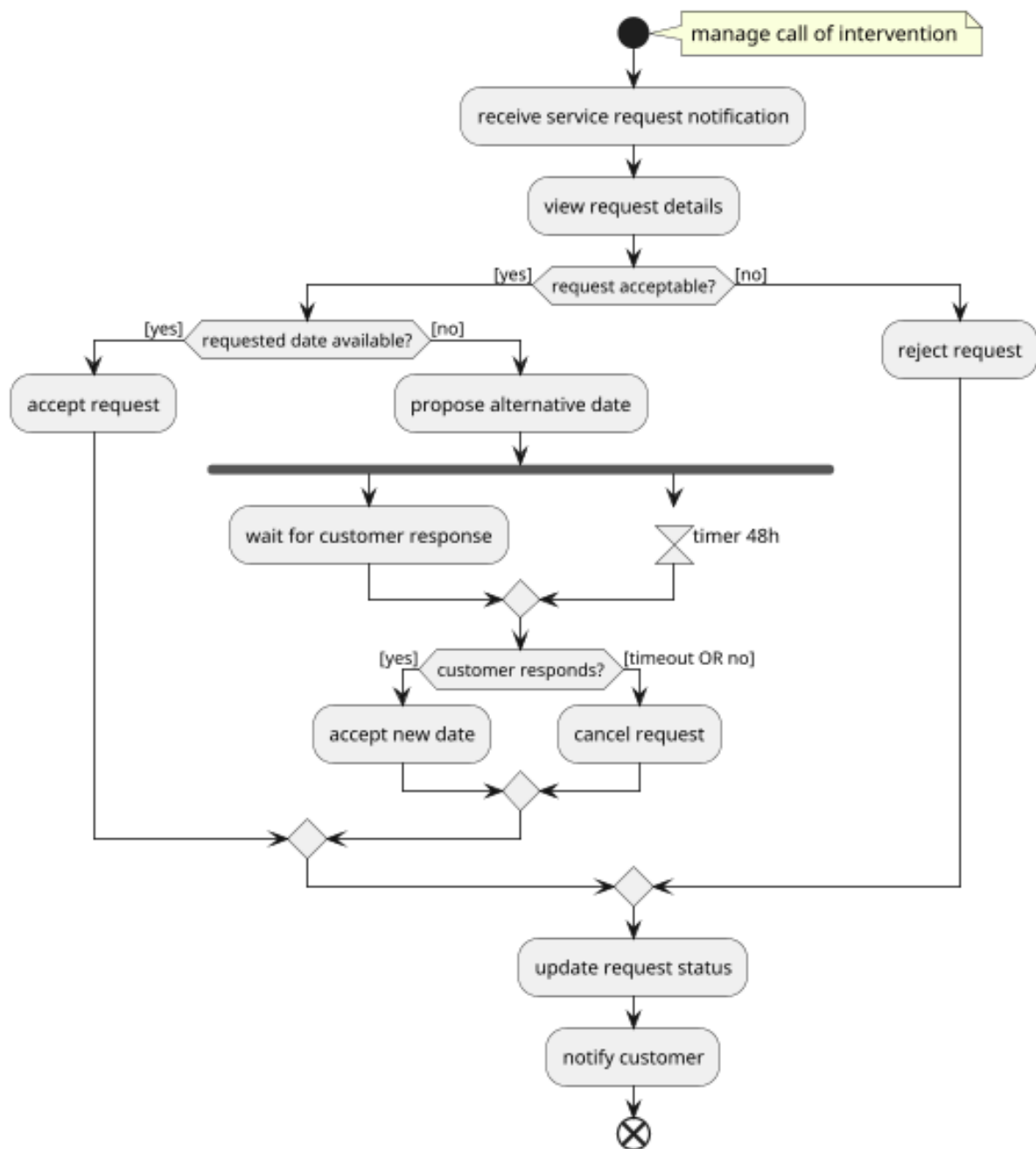
When an application controller updates a booking status or registers a new review, it updates the state within the respective publisher. These publishers manage a dynamic list of subscribers and automatically invoke notifyObservers() whenever a state change occurs, ensuring that all dependent components react seamlessly.

3.3. Activity Diagram

3.3.1. Booking an appointment



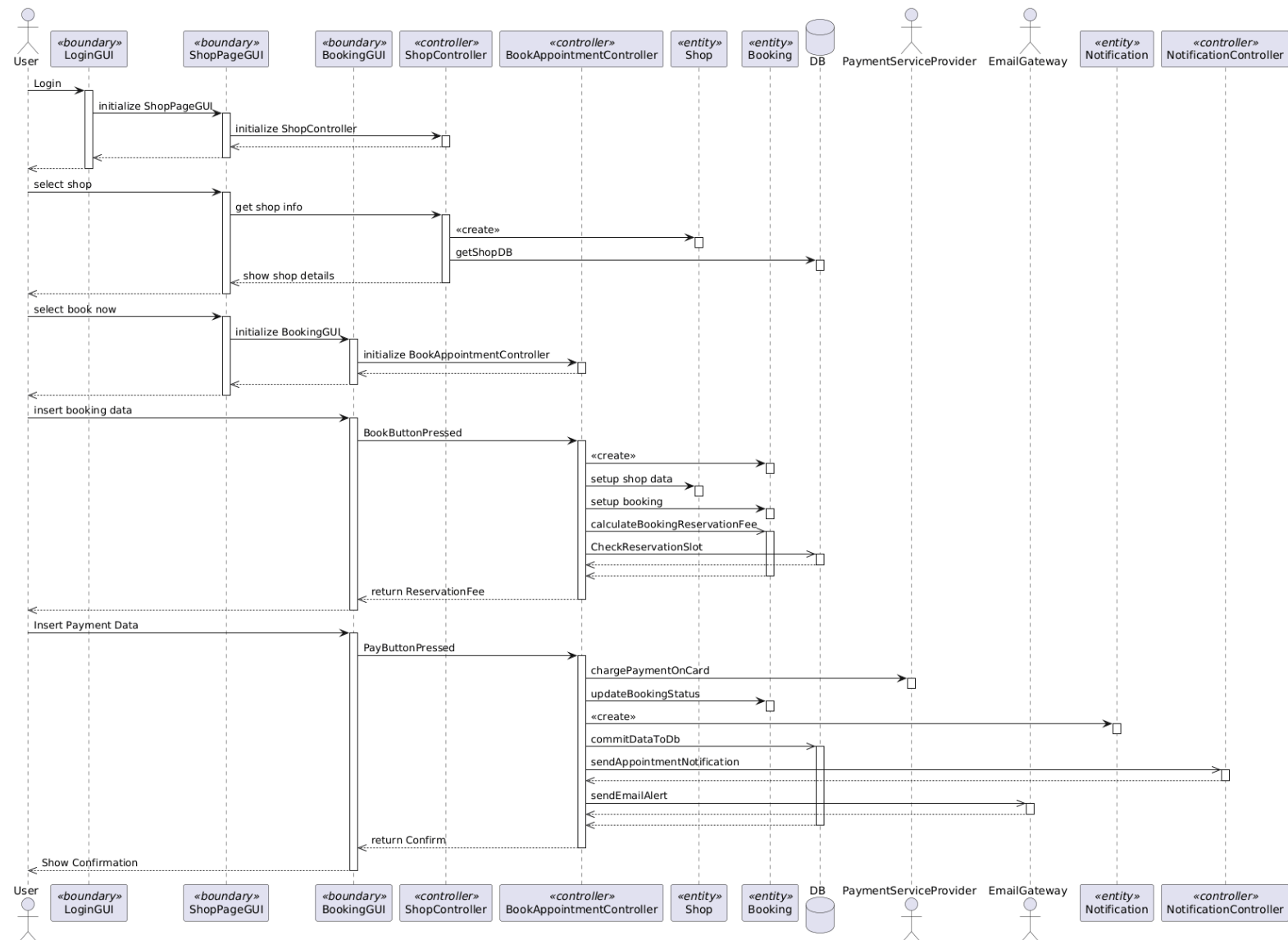
3.3.2. Manage Booking



3.4. Sequence diagram

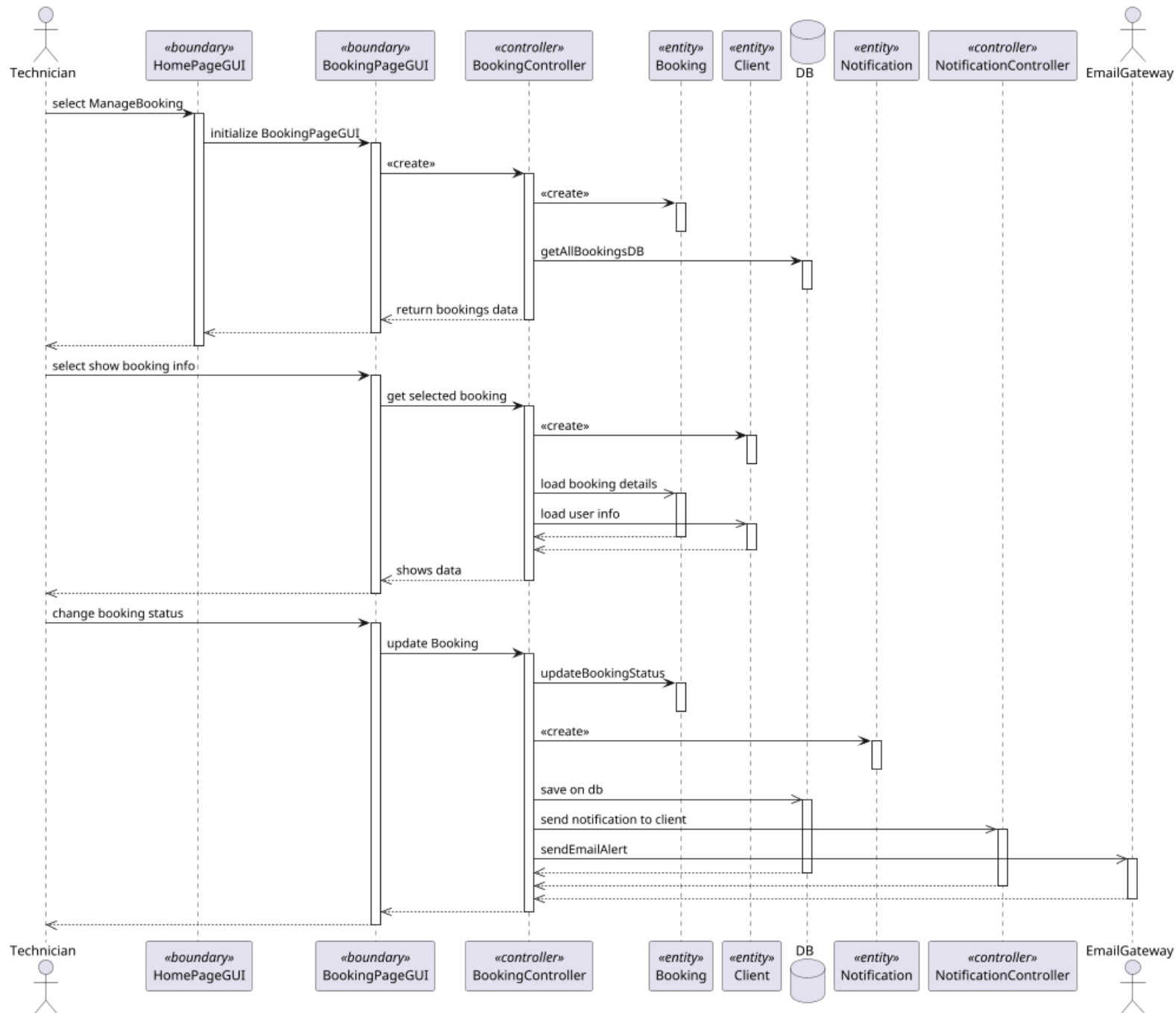
3.4.1. Booking an appointment

Book an appointment



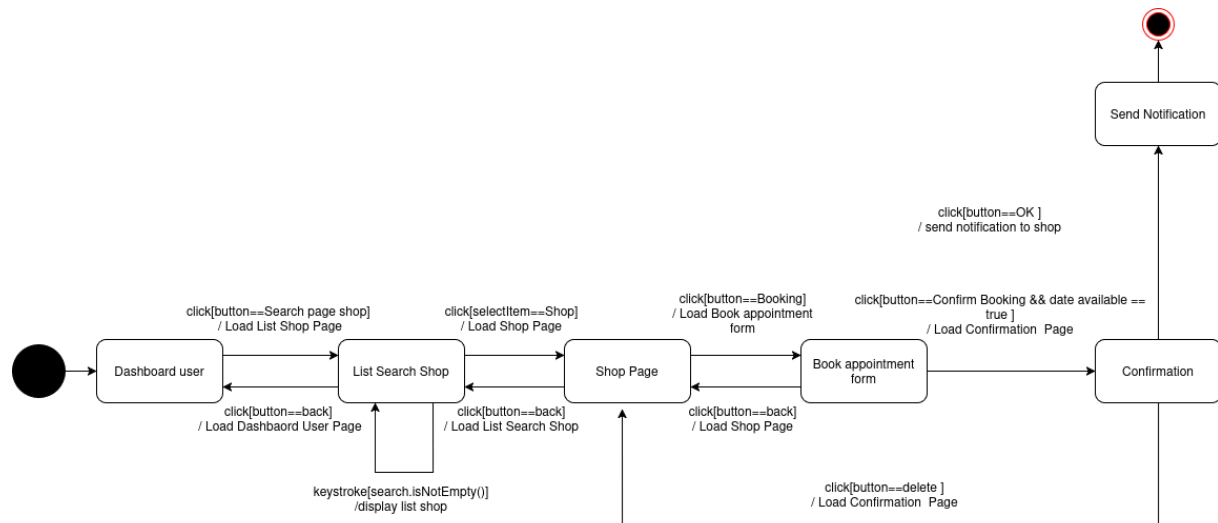
3.4.2. Manage Booking

Manage Booking Status

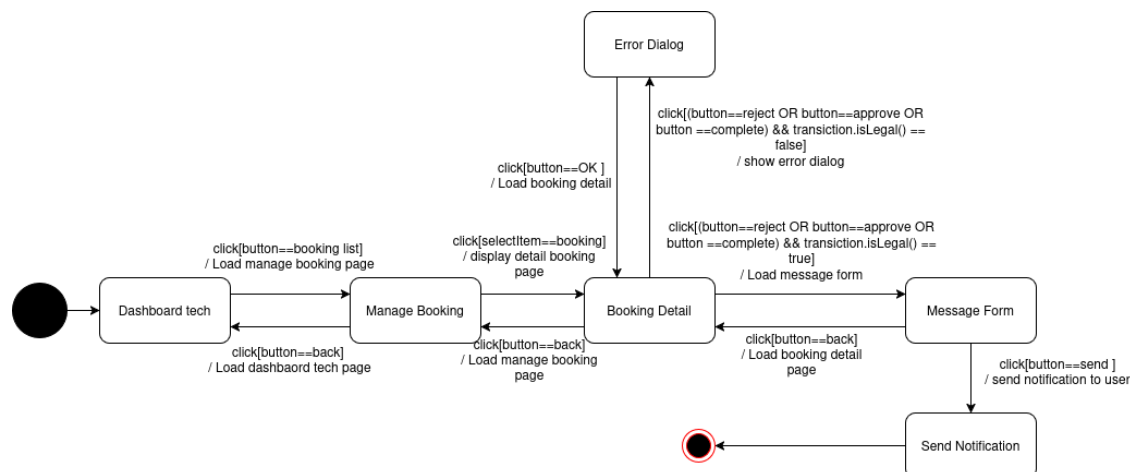


3.5. State diagram

3.5.1. Booking an appointment



3.5.2. Manage Booking



4. Testing

4.1. Booking transition from pending state - Valerio Mazza

We tested that a booking in PENDING state cannot be changed in COMPLETED state. The test verifies that the system prevents invalid state transitions.

4.2. Login with wrong role - Giulio Rustia

We tested login using technician credentials while selecting the CLIENT role and viceversa. The test verifies that authentication fails, ensuring consistency between user identity and role.

4.3. Duplicate booking prevention - Valerio Mazza

We tested that a booking cannot be inserted when the same shop, date, and timeslot already exist. The test verifies an exception is thrown, preventing duplicates.

4.4. Unique username on registration - Giulio Rustia

We tested registration with a username that already exists. The test verifies that the username is reported as unavailable, ensuring account uniqueness.

4.5. Booking payment with unsupported card circuit - Giulio Rustia

We tested creating a booking and finalizing the payment using a card from an unsupported circuit. The test verifies that an exception is thrown, ensuring unsupported payment circuits are correctly rejected.

4.6. Address update for client and shop - Valerio Mazza

We tested address updates for both a client and a technician's shop using an invalid address. The tests verify that an exception is thrown in both cases, enforcing address validation.

5. Code

5.1. GitHub Repository

GitHub JustIT

<https://github.com/GiuEliNo/JustIT.git>

5.2. SonarCloud

SonarCloud JustIT

https://sonarcloud.io/project/overview?id=dosti_justit-prova

6. Video Presentation

YouTube Video

<https://youtu.be/V4Ah5DKxQSU>